

# Multiplying and Dividing Fractions: Find the Flip Mistake

Grade 4–5

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*Cancelling correctly before multiplying, flipping only the divisor, and finding and fixing a misused reciprocal*

**Two rules this page is about.**

**R1. Multiplying:** multiply straight across. You may cancel a *top* number with a *bottom* number first — never a top with a top, and never a bottom with a bottom.

**R2. Dividing:** keep the first fraction, change  $\div$  to  $\times$ , and flip *only the second* fraction (the divisor). Cancel **after** the flip, never before.

For every “find the mistake” problem, name the mistake in words *and* show correct work. Leave answers as fractions (improper fractions are fine).

**1.** Multiply. Cancel a top number with a bottom number first, then multiply straight across.

$$\frac{2}{3} \times \frac{3}{8}$$

Answer: \_\_\_\_\_

**2.** Divide. Write down which fraction you flipped before you compute, and why that is the one that flips.

$$\frac{3}{4} \div \frac{2}{5}$$

Answer: \_\_\_\_\_

**3. Find and fix.** Kai divided like this:

$$\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{4}{5} = \frac{6}{5}$$

(a) Name Kai's mistake in words.      (b) Show the correct work and give the correct answer.

Answer: \_\_\_\_\_

**4. Find and fix.** Maya multiplied like this, saying she "cancelled the 4 and the 8":

$$\frac{4}{5} \times \frac{8}{9} = \frac{1}{5} \times \frac{2}{9} = \frac{2}{45}$$

(a) Explain why that cancelling is not allowed.      (b) Give the correct product.

Answer: \_\_\_\_\_

**5.** Find the missing factor. To undo multiplying by  $\frac{3}{4}$  you divide by  $\frac{3}{4}$ , which means multiplying by its reciprocal  $\frac{4}{3}$ .

$$\frac{3}{4} \times \underline{\hspace{2cm}} = \frac{1}{2}$$

**6. Find and fix.** Jordan cancelled before flipping:

$$\frac{6}{7} \div \frac{3}{8} = \frac{3}{7} \div \frac{3}{4} = \frac{3}{7} \times \frac{4}{3} = \frac{4}{7}$$

(a) Explain why no cancelling is allowed until the division has been rewritten as multiplication. (b) Give the correct answer.

Answer: \_\_\_\_\_

**7. Who is right?** Ali and Sam each computed  $\frac{5}{6} \div \frac{5}{8}$ .

Ali:  $\frac{6}{5} \times \frac{5}{8} = \frac{3}{4}$       Sam:  $\frac{5}{6} \times \frac{8}{5} = \frac{4}{3}$

Which student took the reciprocal of the right fraction? Name the other student's mistake, and write the correct answer.

Answer: \_\_\_\_\_

**8. Find and fix (word problem).** One batch of granola uses  $\frac{3}{4}$  cup of oats. You have  $\frac{9}{2}$  cups of oats. How many whole batches can you make?

Nia wrote  $\frac{2}{9} \times \frac{3}{4} = \frac{1}{6}$  batch. Explain why her answer cannot be right (you have more oats than one batch needs), then give the correct number of batches.

**Answer:** \_\_\_\_\_ batches